Raspberry Pi Pico

+3.3V
R1 220
Reference: 3V1 green
GND
R10 220k
+BATT
GND
+3.3V
J1
5V
NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2
P1
rpi pico 1
2648-SC0915CT-ND
tht
1.8_6.5v
5V
D1 D_Schottky
+3.3V
VBUS
VSYS
3V3_OUT
ADC_VREF
3V3_EN
RUN
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
GP13
GP14
GP15
GP16
GP17
GP18
GP19
GP20
GP21
GP22
GP26
GP27
GP28
AGND
GND
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6
bt1
bt2
bt3
bt4
sda
scl
right_1
right_2
right_en
left_2
left_en
left_1
R2 220
D13
blue
GND
J3 joystick
J4 speed knob

left motor

U1
DRV8245HQPWPRQ1
LVL6, in independent mode that means faults are retried
SR
DIAG
PH/IN2
EN/IN1
DRVOFF
VM
OUT1
OUT2
GND
EPAD
ITIRIP
MODE
NFAULT
IPROPI
NSLEEP
8.2k +-10% LVL2 INDEPENDENT
8.2k R7
left_ipropi
left_en
C1 0.1uF
HSL1 Heatsink
J7 1017491

right motor

U2
DRV8245HQPWPRQ1
right_1
right_2
right_ipropi
right_en
C2 0.1uF
8.2k R6
HSR1 Heatsink
J8 1017491

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR
+3.3V NT1
NetTie_3
TP0 TestPoint
R13 100k
R14 13.7k
right_ipropi
left_ipropi
VDD
SCL
SDA
ADDR
AIN0
AIN1
AIN2
AIN3
NC
C4 0.1uF
TP1 TestPoint
TP2 TestPoint
R11 604
R12 604
R20 EXB34V182JV
pullup resistors
1.8k

qwiic I2C plug
J2 Conn_01x04
MP

weelye compatible header
J11
switch
D2 D_Schottky
bat-
Conn_01x10_Pin

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1 ASJ-200-B-HT
B2 ASJ-200-B-HT
B3 ASJ-200-B-HT
B4 ASJ-200-B-HT

alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593

voltage regulator circuit
U3
TPS54202HDDCR
R5 590k
C6 0.1uF
L1 27 uH
C3 1uF
C7 56pF
R8 100k
R9 13.7k
C8 47uF

battery power circuit
J5 TB005-762-02BE
R3 8.2k
C5 33uf
SW1 screw terminal
18V zener D3
R4 590k
J6
PSMN1R2-25YLDX NMOS Q1
reverse polarity protection
PSMN1R2-25YLDX Q2
on/off switch

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
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